DYSANOS

Generative Dynamic Smooth Arbitrage-free Non-parametric Option Surfaces

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Abstract

This article presents with DYSANOS the first generative market model for smooth SANOS option surfaces for all strikes and expiries which are free of static arbitrage. Our model is designed to generate entire paths of daily spot and option prices for years in the future.

We present a robust and useful if somewhat simplistic baseline in the form of an AR(1) model. We discuss model setup, data pipeline, and training and investigate market reconstruction, stability, and tail behavior. We illustrate model performance on 891 Option Metrics IvyDB S&P Index surfaces from 2022-01-03 through to 2025-08-29.

We also demonstrate how to use the methods of [BMPW22] to construct numerically a risk-neutral density. As part of this we develop a new test for zero conditional means under a given measure. We show that for 100,000 simulated paths a trading universe of 48 options and spot is numerically free of dynamic arbitrage.

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1 Introduction

The implied volatility surface of an option market changes dynamically over time, giving rise to Vega risk when risk managing option portfolios. Modelling the dynamics of a full continuous surface of options in a way that is simultaneously *statistically realistic* and *statically arbitrage-free* at every simulated date has remained an open problem since the foundational work of [CFD02] on dynamics of the implied volatility surface.

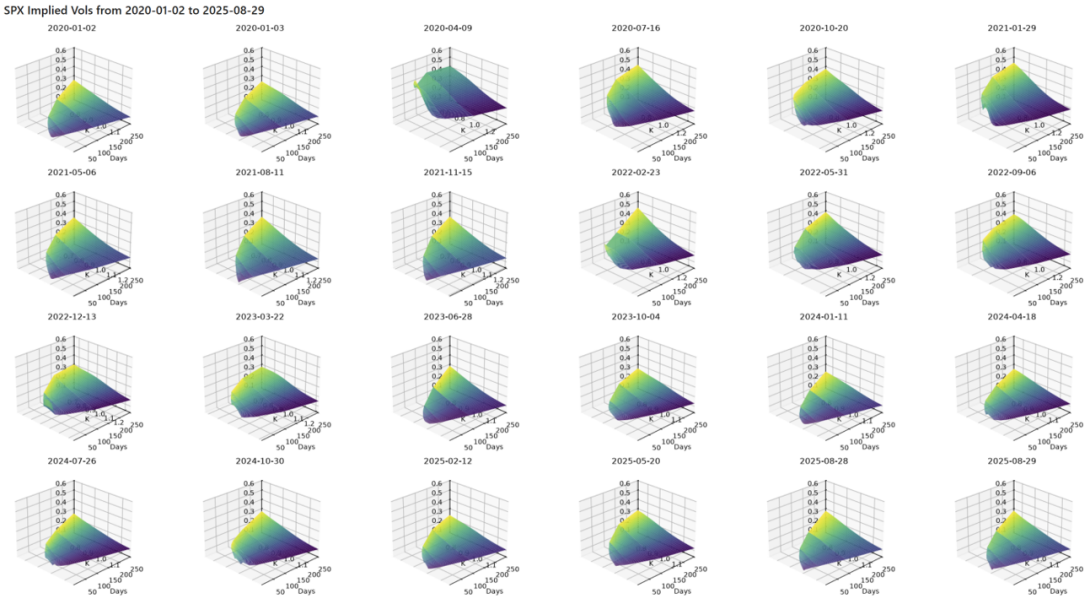


Figure 1: Historical implied volatilities for mid-prices of S&P 500 Index options from 2020-01-02 to 2025-08-29 using Option Metrics Ivy DB data via WRDS `https://wrds-www.wharton.upenn.edu/` (the latest data available at the time of writing).

The most advanced approach until recently were the simulators [WBWB19, WWP+21, CRW23] for realistic dynamics of a *discrete floating grid* of options which are strictly arbitrage-free at each simulation date. These simulators are using, in essence, the discrete local volatility representation pioneered in [BR15] to ensure absence of static arbitrage in every simulated option surface.

However, the restriction of the above models to a discrete set of floating option strikes and time-to-expiries means that the simulators do not have access to the daily returns $dC_t(\tau, k)$ of actual tradable instruments with cash strikes k and calendar expiries τ (because the contract terms change from one simulated day to another). They therefore do not directly provide statistically consistent time series of fixed-contract returns, which are required to validate simulated dynamics against observed market returns. Since the discrete option prices are free of arbitrage, one method would be to interpolate option prices linearly, which means all options would be free of static arbitrage. However, as shown in [Bue06] and illustrated in [BHK+26] linear interpolation yields the most expensive possible arbitrage-free price for any option off the grid, rendering the approach unsuitable in practice. Being linear also means that the implied probability density is piecewise constant and zero between strikes. In other words the spot price can only take values at the model strikes for each expiry to avoid dynamic arbitrage.

This article presents a model framework for

- **(i) realistic generative dynamics** of
- **(ii) smooth option surfaces for all strikes and expiries** which are
- **(iii) free of static arbitrage** and have

- **(iv) a risk-neutral density**.

This article is a refinement of material presented at The XIII Bachelier World Congress in Bologna, 2026.[1]

Historic Context and Literature Review

The classical approach and its limitations: Quantitative finance has traditionally approached this problem by specifying analytically tractable risk-neutral dynamics for the underlying asset — Heston, rough volatility, SABR — and then *calibrating* model parameters to fit each day's option surface independently. They are then overlaid with a local volatility surface using the seminal [GHL12]. The latter guarantees that all options are initially priced correctly.

In this paradigm, realistic surface dynamics are an afterthought: calibration is repeated from scratch on each date. This approach is both impractical for large-scale simulation and structurally inconsistent with the goal of generating realistic implied volatility or option price paths.

Simulating Implied Volatility: A natural data-driven alternative is to treat the implied volatility surface itself as the simulation object, much as implied in [CFD02]. The fundamental difficulty is that there is no known parametrization of the implied volatility surface which is strictly free of static arbitrage. There exist numerous attempts to rectify this with classic (e.g. [Fen09]) and machine learned methods (e.g. [NJZB23]) but to our knowledge none of them can strictly guarantee absence of static arbitrage.

If a simulated surface admits a butterfly spread or calendar spread arbitrage, then a learning agent can exploit an artificial profit which is unrelated to the problem for which the generated market was intended. Aside from those classic constraints call prices also need to adhere to boundary conditions which are often not discussed in the literature (we repeat them below).

Most published works also simplify the problem by only considering few strikes and expiries.

Constrained Parameter Spaces and Code Books: The core challenge of any representation of an option price surface is that no-static-arbitrage conditions are essentially linear conditions on call prices, which are difficult to enforce directly in a generative machine learning model. This challenge is shared by most representations including most parametrizations of implied volatility.

The idea of a "code book" as introduced in [CN12] is to find a representation of the option price surface which is numerically tractable for simulation and for which required parameter constraints are straightforward to enforce. Indeed, the seminal [CN12] achieves that goal. In addition, the authors are able to also construct a risk-neutral measure consistent with the code book representation. They achieve this, essentially, by writing the spot price as a product of independent increments. That construction unfortunately also restricts the dynamics their model can represent. It tends to fit poorly to statistical data due to the strong cross-expiry correlation of implied forward volatilities.

Local Volatility as a Code Book: An intuitive alternative is to use Dupire's local volatility as "code book" instead *(not as dynamical model)*. Given a non-negative local volatility function $\sigma_t(T,K)$ for $(T,K) \in \mathbb{R}^2_{\geq 0}$ defined on some day t we can define a smooth, strictly arbitrage-free option surface $C_t(T,K) := C^{\mathrm{LV}}(\sigma_t; T, K)$ by solving the Dupire forward-PDE $\partial_T C^{\mathrm{LV}}(\sigma_t; T, K) = \frac{1}{2}K^2\sigma_t^2(T,K)\,\partial^2_{KK} C^{\mathrm{LV}}(\sigma_t; T, K)$. That means any statistical generative model of Dupire's local volatility gives rise to a dynamic model of option surfaces which are by construction free of static arbitrage.

[1] https://quantitative-research.de/dl/DYSANOS.pdf

However, in order to price any simulated option off a simulated local volatility surface we would need to solve a forward-PDE on a suitably fine grid which is slow, particularly when used within batch-based gradient descent training.

The idea to address this conundrum is to use a grid of options and define a discrete version of local volatility on that grid, which is what we called *discrete local volatility (DLV)* in [BR15]. It is essentially the discretized local volatility which is consistent with using an implicit forward-PDE operator on a sparse inhomogeneous grid of options, as explained in [AH11]. However, its native option prices for strikes not on its base grid are the result of linear interpolation and therefore, as mentioned above, "too expensive". We therefore use our smooth SANOS extension [BHK+26].

The use of SANOS also allows to fit our model very efficiently to observed bid/ask spreads over the entire observed option surface.

Key contributions

SANOS Decoder: We show how to implement and learn a coordinate transformation of our SANOS [BHK+26] model such that the "ML-SANOS" state parameters h_t live in an unconstrained space $\mathbb{R}^d$. The call prices

$$h_t \longmapsto C(h_t; T, K) \ .$$

remain smooth and statically arbitrage-free.

We show how ML-SANOS can be trained directly to suitably pre-processed option market data, yielding a trained h_t for each day. The pre-processing step means that DYSANOS, via SANOS, is fitted to real market bid/ask prices.

Generative Dynamics: We first show that the direct surface state can be reduced to an affine 200-dimensional representation with little loss of quote fit. We then add a stable one-week AR(1) overlay and jointly refine the surface states and transition. Its transition is parameterized as the matrix exponential of a dissipative generator, and its full innovation covariance is learned by a nonsingular Cholesky factor. The overlay preserves the strong market fit while producing a stationary generative law. That law permits exact Gaussian transitions at daily, one-week, or longer horizons; as non-Gaussian alternatives, we fit the Tail Diffusion, a no-replay scalar log-variance model, and resample historical one-week innovations.

We do not claim that this approach provides the most realistic dynamics; rather, we see this implementation as baseline to beat with more modern time series models.

Risk-Neutral Dynamics: Every decoded SANOS surface is free of static arbitrage by construction. We show that we can "remove the drift" in the sense of [BMPW22] for a subset of the (infinite) tradable instruments by finding a martingale measure $\mathbb{Q}$ equivalent to the learned $\mathbb{P}$. We test the resulting measure against two no-arbitrage criteria: zero unconditional mean and zero conditional expected return. To assess the latter, we develop a novel "no-arb" test. This is of independent interest.

2 SANOS

We consider European call options on a single underlying for historic days $t \in \{t_1, \dots, t_m\} \subset \mathbb{N}$ which we will count in business days. We assume forwards, discount factors, and survival probabilities are deterministic. As described in remark 2.1, we can therefore convert them at each point t to driftless equivalents where spot, forwards, discount factors and survival probabilities are all one, for every t.

At time t, we observe $L_t \in \mathbb{N}$ European call option prices with time-to-expiries $T_t^\ell \geq 0$ and

relative strikes $K_t^\ell \geq 0$ for $\ell = 1, \ldots, L_t$ in the market. Their symbolic payoff is therefore

$$\left(\frac{S_{t+T_t^\ell}}{S_t} - K_t^\ell\right)^+ .$$

We denote by B_t^ℓ and A_t^ℓ their bid and ask prices; mid-price and half-spread are defined (in vector notation) as $C_t := \frac{1}{2}(A_t + B_t)$ and $\gamma_t := \frac{1}{2}(A_t - B_t)$.

We generally assume that $B_t \succ 0$ and $\gamma_t \succ 0$.[2] Contrary to most derivative literature, we do not assume that bid/ask prices are in some way free of arbitrage; instead we will show how to handle arbitrage valuations gracefully.

We will use the Black & Scholes formula in terms of *total volatilities* defined for a spot S, strike K and total volatility W as

$$\text{BSCall}(S, K, W) := S\,\mathcal{N}(d_+) - K\,\mathcal{N}(d_-) \quad \text{for} \quad d_\pm := \frac{1}{W}\log\frac{S}{K} \pm \frac{1}{2}W \tag{1}$$

where $\mathcal{N}$ denotes the normal distribution function.

Remark 2.1 (Conversion to driftless data)**.** *Assume that at each time t spot is $\tilde{S}_t$, forwards for some time to expiry $T \geq 0$ are $\tilde{F}_t(T)$, discount factors are $\tilde{B}_t(T)$. In that case the spot at $t+T$ as seen from time t must be given as*

$$\tilde{S}_T^{(t)} = \tilde{F}_t(T) X_T^{(t)}$$

for a non-negative martingale $X^{(t)}$ with $X_0^{(t)} = 1$.

Assume then that $\tilde{C}_t(T, \tilde{K})$ denotes the market call price for time-to-expiry T and cash strike $\tilde{K}$. Define the relative strike as $K := \tilde{K}/\tilde{F}_t(T)$.

The driftless call price is then given as

$$C_t(T, K) := \frac{\tilde{C}_t\Big(T,\, K\tilde{F}_t(T)\Big)}{\tilde{B}_t(T)\tilde{F}_t(T)} .$$

Puts are defined equivalently. This results can be extended to the presence of cash dividends and deterministic survival probabilities.[3]

Remark 2.2 (Conversion to driftless spot returns)**.** *A consistent driftless spot return series can be obtained by the fully funded one-day forward-return*

$$\frac{S_{t+1}}{S_t} := \frac{\tilde{S}_{t+1}}{\tilde{F}_t(t+1)} . \tag{2}$$

[2] We define $x \succ y \Leftrightarrow \forall i : x^i > y^i$.

[3] If cash dividends are present and if its survival probabilities are deterministic functions $\tilde{Q}_t(T) \equiv \exp(-\lambda_t(T)T) \in [0,1]$ then it was shown in [Bue08] that spot at $t+T$ as seen from time t must be given as $\tilde{S}_T^{(t)} = \big(\tilde{F}_t(T) - \tilde{D}_t(T)\big)X_T^{(t)} + \tilde{D}_t(T)$, conditional on survival. The deterministic function $\tilde{D}_t(T)$ represents floor enforced on the spot price by the presence of future cash dividends, c.f. [Bue08] for specifics.

The driftless strike now becomes $K := (\tilde{K} - \tilde{D}_t(T))/(\tilde{F}_t(T) - \tilde{D}_t(T))$ for which the driftless call price is now

$$C_t(T, K) := \frac{\tilde{C}_t\Big(T,\, K\big(\tilde{F}_t(T) - \tilde{D}_t(T)\big) + \tilde{D}_t(T)\Big)}{\tilde{B}_t(T)\,\tilde{Q}_t(T)\,\big(\tilde{F}_t(T) - \tilde{D}_t(T)\big)} .$$

For puts a similar formula applies after applying put-call parity, see [Bue08].

2.1 Arbitrage-Free Call Price Functions

A call price function is simply a function $C : \mathbb{R}^2_{\geq} \longrightarrow \mathbb{R}_{\geq 0}$ which computes call prices $C(T, K)$ for time-to-expiries $T \geq 0$ and relative strikes $K \geq 0$ (i.e. the call prices are normalized to a spot price of 1). The objective of this section and the next is to generate such functions C_t for each simulated date t such that each C_t is "free of static arbitrage" in the following sense:

Definition 2.3. *A call price surface $C : \mathbb{R}^2_{\geq 0} \to \mathbb{R}_{\geq 0}$ is* free of static arbitrage *if there is no trade which has non-negative returns with a non-zero probability of positive returns.*

This is equivalent to the existence of a martingale Z such that $C(T, K) = \mathbb{E}[(Z_T - K)^+]$.

Theorem 2.4. *A call price surface C is free of static arbitrage if and only if it does not admit negative butterfly or calendar spreads and satisfies the boundary conditions $C(T, 0) = 1$, $C(0, K) = (1 - K)^+$, $\lim_{K \uparrow \infty} C(T, K) = 0$ and $\partial_K C(T, 0) \in [-1, 0)$.*

The stricter condition $\partial_K C(T, 0) = -1$ applies if the stock price cannot become zero.

This has been proven in [Bue06]. Remark 2.5 shows that these conditions each are not superfluous: we give a practical guide on how to construct *actionable* arbitrage strategies if any of the above are violated.

Remark 2.5 (Actionable arbitrage if the conditions of theorem 2.4 are not met)**.** *All of the below are meant as practical heuristics.*

- ***Butterfly arbitrage:*** *this means that the price of a positive butterfly payoff is negative. The arbitrage is to get paid to enter into the butterfly which can only ever generate a positive return.*
- ***Calendar arbitrage:*** *if for the same strike K the call price at T_2 is below the call price at T_1, then:*
  1. *Sell the call at T_1 and buy the call at T_2. This provides a positive cash flow $\pi_0 > 0$.*
  2. *In T_1 if the call expires out of the money, then we have made money: both π_0 and a potential positive payoff at T_2 in our favour.*
     If the call expires in the money, then we deliver one share and receive K. We are now short that share. In T_2, if the call expires in the money, then we receive one share to cover our short in return for K. Our total cashflow is π_0. If the call expires out of the money, then $S_T < K$. We can therefore buy back the share and cover our short for less than K. Our profits are $\pi_0 + (K - S_T)^+ > 0$.
- *If $C(0, K) \neq (1 - K)^+$ we have an immediate pathological arbitrage.*
- *If $C(T, 0) > 1$ then sell the call and cover with one share; if $C(T, 0) < 1$ then buy the call and enter into a short equity position.*
- *$\partial_K C(T, 0) \geq -1$: assume on the contrary there is a $K \downarrow 0$ such that $(1 - C(T, K))/(-K) < -1$ i.e. $C(T, K) = 1 - K - \epsilon$ for some $\epsilon > 0$. Buy the call and short sell the equity, resulting a cash position of $\pi_0 = -K + \epsilon$; the payoff in T is $(S_T - K)^+$ which allows us to cover our position and cash with a total reward of $\epsilon > 0$.*
- *$\partial_K C(T, 0) < 0$: assume on the contrary there is a $K \downarrow 0$ such that $(1 - C(T, K))/(-K) \geq 0$. We show existence of arbitrage in the case of equality i.e. existence of some $K > 0$ such that $C(T, K) = 1$. In that case we sell the call and buy the equity for 1. If the call expires in the money, we deliver the share and receive $K > 0$. If the call expires out of the money, then sell the share for $S_T \geq 0$.*

- $\lim_{K\uparrow\infty} C(T,K) = 0$ *which means for all* $c > 0$ *there exists some* K *such that* $C(T,K') < c$ *for all* $K' \geq K$ *(because of convexity). This is the only arbitrage condition which is technical because a violation means, essentially, that the equity can grow without bounds which violates integrability. Assume therefore on the contrary that there is some* $c > 0$ *such that* $C(T,K) \geq c$ *for all* K*. Notice that, formally,*

$$S_T = -\int_0^\infty \frac{(S_T - K + dK)^+ - (S_T - K)^+}{dK} dK = S_T + \lim_{K\uparrow\infty} (S_T - K)^+ .$$

Therefore sell (a discretization of) the strip

$$\int_0^\infty \frac{C(T,K) - C(T,K+dK)}{dK} dK = 1 + e$$

which delivers S_T*, and buy the equity. This will generate profits of* $e > 0$*.*

2.2 Smooth Arbitrage-free Non-parametric Option Surfaces

In [BHK+26] we introduced a non-parametric family of call price functions which are all statically arbitrage-free. For clarity, let us briefly fix t and drop it from the notation of market data and model parameters.

Each call price function is given in terms of model expiries $0 = \hat{T}^0 < \hat{T}^1 < \cdots < \hat{T}^M$ and, for each model expiry, $N^j > 0$ inner strikes.[4] Including the two boundary strikes, write $\hat{K}^j = (\hat{K}^{j,i})_{i=0,\ldots,N^j+1}$. It is recommended that the two outer strikes are far away from the inner strikes.[5] Model strikes and expiries do not need to match market strikes or expiries. See [BHK+26] for general comments on grid construction.

We say that $q = (q^1, \ldots, q^M)$ with $q^j \in \mathbb{R}^{N^j+2}$ is a *martingale density* if it satisfies the linear constraints

$$\begin{cases} \text{Each } q^j \text{ is a density:} & q^j \geq 0,\ (q^j)'1 = 1 \\ \text{Each } q^j \text{ has unit expectation:} & (q^j)'\hat{K}^j = 1 \\ \text{The } q \text{ are in convex order:} & q^{j\prime} CX_j \geq q^{j-1\prime} CX_{j|j-1} \end{cases} \tag{3}$$

The matrices CX_j and $CX_{j|j-1}$ project the density on call prices. They are defined as

$$CX_j^{i,k} := (\hat{K}^{j,i} - \hat{K}^{j,k})^+ \quad i = 0, \ldots, N^j + 1;\ k = 1, \ldots, N^j$$

and

$$CX_{j|j-1}^{i,k} := (\hat{K}^{j-1,i} - \hat{K}^{j,k})^+ \quad i = 0, \ldots, N^{j-1} + 1;\ k = 1, \ldots, N^j .$$

Accordingly, $q^{j\prime} CX_j \in \mathbb{R}^{N_j}$ is the vector of call prices for strikes $\hat{K}^j \in \mathbb{R}^{N_j}$ computed using the density q^j while $q^{j-1\prime} CX_{j|j-1} \in \mathbb{R}^{N_j}$ is the vector of call prices for the same strikes $\hat{K}^j \in \mathbb{R}^{N_j}$ computed using the previous density q^{j-1}. The condition $q^{j\prime} CX_j \geq q^{j-1\prime} CX_{j|j-1}$ enforces that call prices are increasing into each strike $\hat{K}^j$.

Definition 2.6 (SANOS call price)**.** *Let* q *be a martingale density and* $0 < W^1 < \cdots < W^M$ *be increasing smoothing total volatilities.*[6] *The SANOS call price for arbitrary* $T \geq 0, K \geq 0$ *is given by*

$$\begin{aligned} C(T,K) := \alpha^j(T) \sum_{i=0}^{N^j+1} \mathrm{BSCall}(\hat{K}^{j,i}, K, W(T)) q^{j,i} \\ + (1 - \alpha^j(T)) \sum_{i=0}^{N^{j-1}+1} \mathrm{BSCall}(\hat{K}^{j-1,i}, K, W(T)) q^{j-1,i}, \qquad T \in (\hat{T}^{j-1}, \hat{T}^j]. \end{aligned} \tag{4}$$

[4] Note that we changed notation slightly vs. [BHK+26].

[5] In our experiments, the inner strikes $0 < \hat{K}^{j,1} < \cdots < 1 < \cdots < \hat{K}^{j,N^j}$ were chosen first and then extended by $\hat{K}^{j,0} := \hat{K}^{j,1}/2$ and $\hat{K}^{j,N^j+1} = 1.5\,\hat{K}^{j,N^j}$.

[6] The linear-programming preprocessing uses 50% of ATM total volatility. The trained ML-SANOS decoder instead learns the positive forward smoothing multipliers η^j.

with $\alpha^j(T) = (T - \hat{T}^{j-1})/(\hat{T}^j - \hat{T}^{j-1})$ *and total volatility* $W(T) := \alpha^j(T)W^j + (1 - \alpha^j(T))W^{j-1}$.

For $T > \hat{T}^M$ *we set* $\alpha^{M+1}(T) = 0$ *and* $W(T) := W^M\sqrt{T/\hat{T}^M}$*; however extrapolation beyond the last model expiry or before the front expiry is generally not recommended.*

Theorem 2.7. *The call price function* C *is statically arbitrage-free. For* $W(T) > 0$ *it is* C^∞ *in* K*. The function is as smooth as* α *in* T*.*

Remark 2.8. *Contrary to [BHK+26] we interpolate the Black & Scholes backbone in total volatility linearly, not variance. This is a mechanical choice to avoid taking square roots of variables during training which can lead to exploding gradients. If forward volatilities are bounded well away from zero, linear interpolation in variance is suitable, too.*

Remark 2.9. *The Black & Scholes formula in (4) can be replaced by the call price function for any martingale in* T*, for example the FFT Heston price formula [CM99]. For practical implementations, the function should be fast on a GPU.*

Calibration as a linear program: We assume we obtain $W^1 < \cdots < W^M$ from the market e.g. as square roots of ATM variance per expiry. Given W the model call price $C^\ell := C(T^\ell, K^\ell)$ for a market expiry T^ℓ and a market strike K^ℓ can be seen in (4) to be linear in q. The market-fitting problem can therefore be formulated as the linear programme

$$\min_q \sum_\ell \frac{(C^\ell - \boldsymbol{A}^\ell)^+ + (\boldsymbol{B}^\ell - C^\ell)^+ + \epsilon|C^\ell - \boldsymbol{M}^\ell|}{(A^\ell - B^\ell)/2} \tag{5}$$

subject to the linear constraints (3) encoding non-negativity, normalisation, integration to 1, and the inter-expiry convex-order constraints that enforce absence of calendar-spread arbitrage. We note that the model strikes and time-to-expiries do not need to correspond to market strikes or expiries.

As discussed in [BHK+26] the primary application of this model is *interpolation*: if the model strikes are a superset of *all* strikes of relevant market instruments then the model can practically be fit to the vast majority of options across the entire surface from 1 day to expiries to several years. Stronger, if the model is linear ($W = 0$) the existence of the densities q is equivalent to absence of arbitrage with respect to the market bid/asks (usually this simply means our market data is not aligned correctly).

On S&P 500 Index data with 1000 options across 48 expiries (2-day to 657 business-day TTE), the LP solves in sub-seconds on a desktop PC, achieving 91.4% of options within bid–ask on a representative date (2025-05-06), with a median error of 21% of the half-spread on the remaining options. Figure 2 illustrates the quality of the fit. The example is from [BHK+26]; refer to that work for further details.

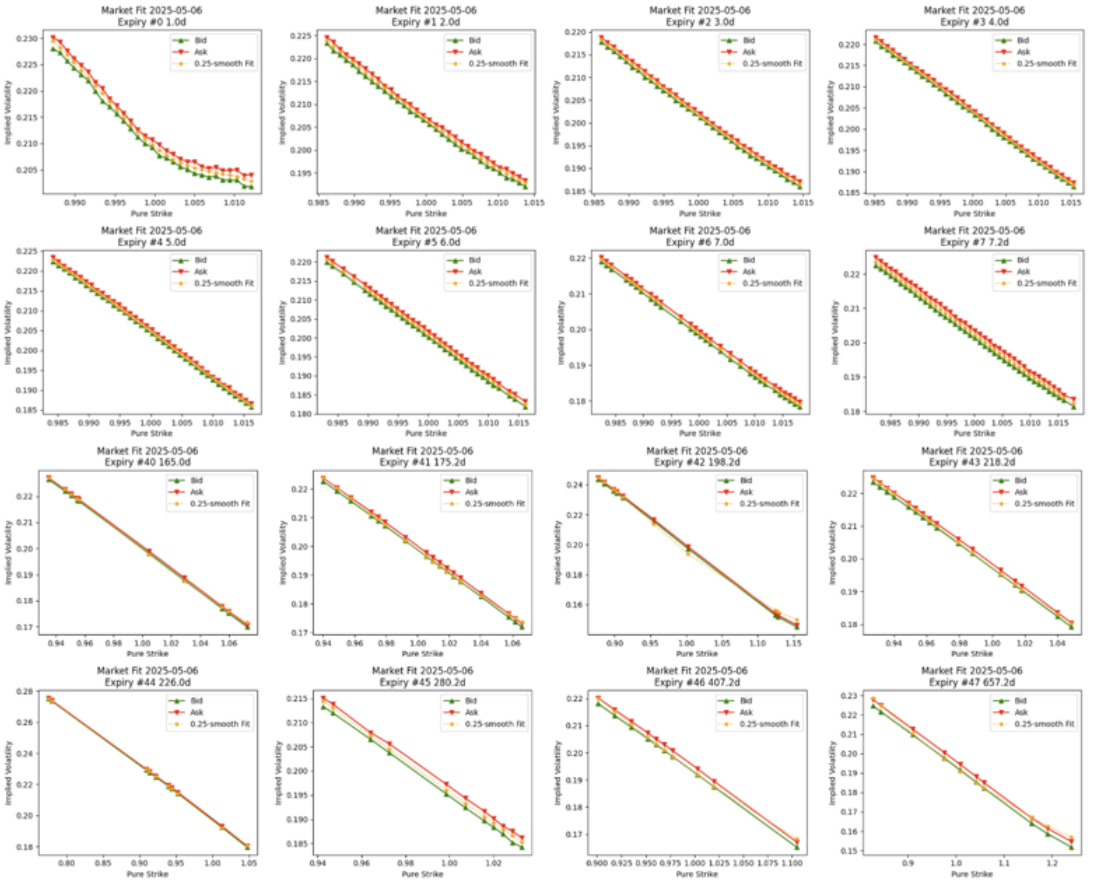


Figure 2: Example fit of SANOS.

2.3 ML-SANOS

We may view our SANOS call price operator as a map from the model parameters to a function space

$$(q, W; \hat{K}, \hat{T}) \overset{\text{SANOS}}{\longmapsto} C \tag{6}$$

for which we naturally want to learn q, W and, as we will see, $\hat{K}$ (as the right strike grid depends on the implied variance of the market). However, the density weights q^j in particular are subject to the linear constraints (3) for which no exact methods exist to integrate them into standard batch gradient descent methods. In this section we therefore reparameterize the map (6) in terms of learnable parameters.

To do so, we no longer require strict *interpolation* of the market prices. While we will want to be as close as possible to observed market prices (and will show that we are able to do so), we now focus on a model parametrization which lends itself to efficient learning using standard optimization techniques. That means that the number of strikes must somehow be kept manageable, rather than be dictated by the number of market instruments at each observed historic day t.

As a first step we therefore fix a common strike grid $\hat{K}$ with $0 < \hat{K}^0 < \cdots < 1 < \cdots < \hat{K}^{N+1}$ across expiries to simplify our model. The call price for strike $\hat{K}^i$ given the density q^j is given as $c^{j,i} = (q^j)'(\hat{K} - \hat{K}^i)^+$. The convex order condition in (3) therefore becomes simply $c^{j,i} \geq c^{j-1,i}$ for all i.

We now reparameterize our densities in terms of *discrete local volatilities (DLVs)* [BR15]: let

$$\omega^{i\pm} := \pm \frac{(\hat{K}^i)^2}{(\hat{K}^{i+1} - \hat{K}^{i-1})\,(\hat{K}^{i\pm1} - \hat{K}^i)} \geq 0$$

and with it the tridiagonal matrix Ω with

- Upper diagonal $(-\omega^{1-}, \dots, -\omega^{N-}, 0)$,
- Lower diagonal $(0, -\omega^{1+}, \dots, -\omega^{N+})$, and
- Main diagonal $(0, +\omega^{1+} + \omega^{1-}, \dots, +\omega^{N+} + \omega^{N-}, 0)$.

i.e.

$$\Omega = \begin{pmatrix} 0 & -\omega^{1-} & 0 & & & & \\ 0 & +\omega^{1+} + \omega^{1-} & -\omega^{2-} & & & & \\ 0 & -\omega^{1+} & +\omega^{2+} + \omega^{2-} & & & & \\ & & & \ddots & & & \\ & & & \ddots & & & \\ & & & & +\omega^{(N-1)+} + \omega^{(N-1)-} & -\omega^{N-} & 0 \\ & & & & -\omega^{(N-1)+} & +\omega^{N+} + \omega^{N-} & 0 \\ & & & & 0 & -\omega^{N+} & \end{pmatrix}. \tag{7}$$

Then for *any* squared (discrete) local volatilities $\Sigma^j = (0, \Sigma^{j,1}, \dots, \Sigma^{j,N}, 0)$ the operator

$$1 + \Omega \operatorname{diag}(\Sigma^j)(\hat{T}_j - \hat{T}_{j-1}) \in \mathbb{R}^{(N+2)\times(N+2)} \tag{8}$$

is a tridiagonal operator with positive main diagonal and negative off diagonals. It is basically the discrete implicit scheme for the forward equation of the continuous time Dupire SDE. It is well known [BR15] that this operator is invertible, and the inverse Ξ^j can be efficiently computed on GPU using a compiled Thomas' algorithm. It is also a martingale transition operator in the sense of [BR15] which means that the densities $q^1, \dots, q^M$ generated iteratively starting in $q^0 = (1)$ by

$$q^j = \left(1 + \Omega \operatorname{diag}(\Sigma^j)(\hat{T}_j - \hat{T}_{j-1})\right)^{-1} q^{j-1} \tag{9}$$

form a martingale density over $\hat{K}$.

With the above we have established a map from much simpler non-negative discrete local vols squared to densities, and then to SANOS call prices:

$$(\Sigma, W;\, \hat{K}, \hat{T}) \overset{\text{DLV}}{\longmapsto} (q, W;\, \hat{K}, \hat{T}) \overset{\text{SANOS}}{\longmapsto} C\,. \tag{10}$$

That allows us to pass to a practically unconstrained parameter set, which we call the ML-SANOS parametrization. We will use the sigmoid function $\operatorname{sigmoid}(x) = (1 + e^{-x})^{-1}$.

Definition 2.10 (ML-SANOS)**.** *Fix the normalized grid geometry for ML-SANOS with:*

- *Time-to-expiries (TTEs)* $0 = \hat{T}^0 < \dots < \hat{T}^M$ *with* $\Delta\hat{T}^j = \hat{T}^j - \hat{T}^{j-1}$*;*
- *Normalized strikes* $z^1 < \dots < 0 < \dots < z^N$ *which we assume contain zero;*
- *Minimum and maximum forward volatilities* $\sigma_{\max} > \sigma_{\min} > 0$*;*
- *Minimum and maximum squared discrete local volatilities* $\Sigma_{\max} > \Sigma_{\min} > 0$*; and*
- *Minimum and maximum smoothing factors* $1 > \eta_{\max} > \eta_{\min} \geq 0$*.*

We translate the new real-valued ML-SANOS parameters

$$u = (u^\Sigma, u^\sigma, u^\eta) \in \mathbb{R}^{M\times N+M+2} \tag{11}$$

into SANOS parameters as follows:

- *Total volatilities: define the* ***forward volatilities***

$$\sigma^j := \sigma_{\min} + \text{sigmoid}(u^{\sigma,j})(\sigma_{\max} - \sigma_{\min}) \tag{12}$$

and two smoothing factors

$$\eta^{\text{f}} := \eta_{\min} + (\eta_{\max} - \eta_{\min})\text{sigmoid}(u^{\eta,1}) \quad \textit{and} \quad \eta^{\text{term}} := \eta_{\min} + (\eta_{\max} - \eta_{\min})\text{sigmoid}(u^{\eta,2}) \ . \tag{13}$$

With these quantities define the ***total volatilities***

$$\widetilde{W}^j := \sum_{\ell=1}^{j} \sigma^\ell \sqrt{\Delta\hat{T}^\ell} \qquad \textit{and} \qquad W^j := \eta^{\text{f}}\sigma^1\sqrt{\Delta\hat{T}^1} \ + \eta^{\text{term}} \sum_{\ell=2}^{j} \sigma^\ell \sqrt{\Delta\hat{T}^\ell} \ . \tag{14}$$

- *We use* $\widetilde{W}^j$ *to define a fixed grid as* ***relative strikes***

$$\hat{K} := \exp\left(z\widetilde{W}^1\right) \bigcup \exp\left(z^i\widetilde{W}^M\right) \ , \tag{15}$$

sufficiently sorted, with boundaries $\hat{K}^0 := \frac{1}{2}\hat{K}^1$ *and* $\hat{K}^{N+1} := \frac{3}{2}\hat{K}^N$.

- ***Squared discrete local volatilities****:*

$$\Sigma^{j,i} := \min\Big\{\Sigma_{\max}, \Sigma_{\min} + \exp(u^{\Sigma,j,i})\Big\} \ . \tag{16}$$

(This formulation avoids saturation of the sigmoid function as the raw log-DLV coordinate approaches the cap $\Sigma_{\max}$*; it requires adding a penalty term to the loss function.)*

In our experiments we use the bounds $(\Sigma_{\min}, \Sigma_{\max}) = (0.01^2, 3^2)$, $(\sigma_{\min}, \sigma_{\max}) = (0.01, 1)$, and $(\eta_{\min}, \eta_{\max}) = (0.01, 0.50)$.

Put together, this process describes a map

$$(\,u; z, \hat{T}\,) \overset{\substack{\text{ML}\\ \text{SANOS}}}{\longmapsto} (\,\Sigma, W; \hat{K}, \hat{T}) \overset{\text{DLV}}{\longmapsto} (\,q, W; \hat{K}, \hat{T}) \overset{\text{SANOS}}{\longmapsto} C \ . \tag{17}$$

In our new parametrization u lives in an unrestricted space and is therefore particularly suited for ML/AI based learning. The expiry locations remain fixed.

3 DYSANOS

Given market data for $t = t_1, \ldots, t_m$ we now aim to estimate a dynamic model for the parameters (11) for each day,

$$u_t = \left(u_t^\Sigma,\, u_t^\sigma,\, u_t^\eta\right) \in \mathbb{R}^{M(N+1)+2}. \tag{18}$$

We use $M = 18$ model expiries given by

$$(\ 1, 2, 3, 4, 5, 10, 15, \ldots, 70\ )$$

business days. We use $N = 200$ strikes computed using (15) for z strikes within $(-3, 2)$. The model therefore has 3,620 native coordinates.

Data and model geometries: The empirical population sourced from WRDS' Option Metrics Ivy data base contains 891 S&P Index dates from 2022-01-03 through 2025-08-29. See appendix E for our baseline pipeline we use for all our research with this data set. In addition to the filters described there we here also exclude options outside the model's strike range (which is effectively

$(-3, 2)$ NMS) and any options expiring past the last model expiry of 70 business days. We also exclude tomorrow's AM-settled options.

We obtain 3,566,258 original call-or-put book observations for our model. All options are represented in call-equivalent units through put–call parity; call and put books at the same expiry and strike remain separate observations and contribute separate loss terms.

Economic Objective: Our economic objective to minimize the normalized *excess* outside the bid-ask spread, defined here as

$$e(C; B, A) := \frac{(C - A)^+ + (B - C)^+}{(A - B)/2}. \tag{19}$$

We report the percentage of options inside the bid-ask spread ($e = 0$) and also the mean and conditional excess conditional on not being within bid-ask.

That means from economic perspective our objective is to minimize the normalized excess outside the bid-ask spread for each quote plus a small additional regularization term to pull the solution towards the mid. Concretely, for the ℓth quote on date t let C_t^ℓ be the model call price, A_t^ℓ the ask price, B_t^ℓ the bid price, and $M_t^\ell = (\ A_t^\ell + B_t^\ell)/2$ the mid price.

Then define our loss as

$$\mathcal{L}_t^{\text{quote}} := \frac{1}{L_t} \sum_{\ell=1}^{L_t} \frac{(C_t^\ell - A_t^\ell)^+ + (B_t^\ell - C_t^\ell)^+ + 0.05|C_t^\ell - M_t^\ell|}{(A_t^\ell - B_t^\ell)/2}. \tag{20}$$

3.1 Affine Compression

Considering the large state space we employ an affine compression to reduce dimensionality while retaining most of the relevant information. This "affine-200" representation is defined by the decoder

$$u_t = b + Bh_t \tag{21}$$

in terms of learned $h_t \in \mathbb{R}^{200}$, $b \in \mathbb{R}^{M(N+1)+2}$ and $B \in \mathbb{R}^{(M(N+1)+2)\times 200}$.[7] Table 1 first shows that dimension reduction mostly works: the affine-200 variant costs only 1.36 percentage points in bid-ask fit.

Geometry	Inside bid–ask	Mean excess	Cond. mean	Cond. median
Full	82.28%	0.31	1.76	0.69
Affine-200	80.92%	0.32	1.69	0.68

Table 1: Aggregate fit of the affine-200 vs the full model, measured by *excess* (19).

However, the fit for the one-day to expiry range in the affine-200 model is noticeably worse, highlighting the challenge of capturing very short-term dynamics.

Geometry	1D	2D	3D	4D	5D
Full	94.56%	95.21%	95.57%	95.77%	68.43%
Affine-200	88.59%	91.98%	93.68%	94.13%	67.58%

Table 2: Inside-spread rates at the five shortest nominal expiries.

[7] The choice of 200 is an empirical capacity compromise rather than a structural claim about the intrinsic dimension of option surfaces. It is large enough to preserve almost all of the direct native quote fit while remaining below the number of historical dates, so a full innovation covariance and stable joint state dynamics can still be estimated. We report the native-versus-200 fit loss explicitly below rather than treating this dimension as exact.

3.2 Linear Mean-Reverting State-Space Model

In the previous section we learned per date ML-SANOS parameters u_t for each $t = 1, \ldots, t_m$. The next step is to learn a dynamic model. Since we have shown that we can represent the full u_t sufficiently well using an affine-200 reduction, we will focus on that setup.

Our generative specification learns jointly the affine representation decoder and a stable linear state process: for $d = 200$ let

$$u_t = b + Bh_t, \qquad h_t \in \mathbb{R}^d, \tag{22}$$

$$h_{t+\Delta} = m + A_\Delta(h_t - m) + G'_\Delta z_{t,\Delta}, \qquad A_\Delta = \exp(-\kappa\Delta), \qquad z_{t,\Delta} \sim \mathcal{N}(0, I_d). \tag{23}$$

Thus the innovation covariance at horizon Δ is $G'_\Delta G_\Delta$. Here b, B are the affine projections and A_Δ is a mean-reversion transition matrix. Mean reversion is imposed structurally through

$$S := \tfrac{1}{2}(R - R'), \qquad S' = -S, \qquad \kappa = LL' + S + \kappa_{\min} I, \qquad \kappa_{\min} = 0.0002 \text{ per trading session.} \tag{24}$$

Its symmetric part is $LL' + \kappa_{\min} I \succ 0$. Therefore every eigenvalue λ of κ satisfies $\Re\lambda \geq \kappa_{\min} > 0$, and every eigenvalue of A_Δ has modulus $\exp(-\Delta\Re\lambda) < 1$. Hence the transition is stable for every $\Delta > 0$. This restriction holds during every optimizer update.

The transition likelihood is fitted using every pair of observed surfaces separated by exactly $\Delta = 5$ NYSE trading sessions. We choose this one-week estimation horizon as a stable baseline: it attenuates the influence of noisy one-day surface moves while retaining 861 overlapping transitions for estimation. Daily simulation uses the same generator through $A_1 = \exp(-\kappa)$ and factors the corresponding one-session innovation covariance as $G'_1 G_1$; it is not a separately fitted daily AR model. At the fitted horizon, G_5 is a full upper-triangular matrix whose diagonal is a positive floor plus a softplus transform. Consequently $G'_5 G_5$ is positive definite throughout training.

Let $D = 891$ and let $\mathcal{W}$ contain the 861 exact one-week date pairs. For $(t, s) \in \mathcal{W}$ with $s = t + \Delta$, set

$$e_{t,s} := h_s - m - A_\Delta(h_t - m). \tag{25}$$

The Gaussian transition loss per state dimension is

$$\mathcal{L}_{\mathrm{AR}} := \frac{1}{2d}\left\{\frac{1}{|\mathcal{W}|}\sum_{(t,s)\in\mathcal{W}}\left\|(G'_5)^{-1}e_{t,s}\right\|_2^2 + 2\sum_{i=1}^{d}\log(G_5)_{ii} + d\log(2\pi)\right\}. \tag{26}$$

With the smooth DLV-cap penalty $\mathcal{L}_{\mathrm{cap}}$ and affine-gauge penalty $\mathcal{L}_{\mathrm{gauge}}$ defined in Appendix B, the jointly trained affine state-space model uses

$$\mathcal{J}_{\mathrm{joint}} := \frac{1}{D}\sum_{t=1}^{D}\mathcal{L}_t^{\mathrm{quote}}(b + Bh_t) + \lambda_{\mathrm{cap}}\mathcal{L}_{\mathrm{cap}} + \lambda_{\mathrm{AR}}\mathcal{L}_{\mathrm{AR}} + \lambda_{\mathrm{gauge}}\mathcal{L}_{\mathrm{gauge}}, \tag{27}$$

where

$$\lambda_{\mathrm{cap}} = 1, \qquad \lambda_{\mathrm{AR}} = 0.10, \qquad \lambda_{\mathrm{gauge}} = 0.10.$$

The model is deliberately simple. The affine decoder reduces a 3,620-dimensional arbitrage-free surface coordinate to a state small enough to support a nonsingular full innovation covariance. The matrix exponential supplies stable transitions at every simulation horizon, while joint estimation chooses a representation that remains accurate in quote space and is also statistically coherent through time.

The implementation minimizes (27) over 300 epochs by alternating surface and transition updates, with the AR weight ramped from zero to its stated value over the first 50 epochs. Table 3 shows that the expanded estimation does not impede overall fit.

Geometry and state	Inside bid–ask	Mean excess	Cond. mean	Cond. median
Full	82.28%	0.31	1.76	0.69
Affine-200	80.92%	0.32	1.69	0.68
AR(1) affine-200	**81.48%**	0.32	1.72	0.68

Table 3: Aggregate fit before and after the joint AR overlay in terms of *excess* (19).

Remark 3.1. *Gaussian innovations miss most of the tail events present in the empirical data. See appendix C on page 28 for capturing tail dynamics better.*

Remark 3.2. *The 80.92% and 81.48% market-fit figures were computed in-sample on the population used to estimate the final all-history model because the aim here was to build a model which is self-consistent within this population.*

It is not a claim that b and B will maintain the same market-fit out-of-sample. A future forecasting study should freeze the decoder and dynamics before the evaluation period.

3.2.1 Spot

The normalized floating option surface can be simulated without considering spot. However, for most applications we will still need a spot path. We will essentially aim to draw spot from the front expiry marginal distribution. This means that generated spot paths will be consistent with the one day to expiry option prices. In order to ensure a correct joint distribution between spot and the surface, we use a "copula" approach.

Assume that we have historic samples of daily driftless spot returns S_{t+1}/S_t as defined in (2). Let F_t^1 denote the CDF obtained[8] from the front one-day decoded SANOS density q_t^1 and use it to transform the spot return into the Gaussian score

$$dg_t = \Phi^{-1}\left[F_t^1\left(\frac{S_{t+1}}{S_t}\right)\right]. \tag{28}$$

We now simply regress dg_t on the normal $z_{t,1}$ from (23). If $\sigma_\perp$ is the fitted residual standard deviation, we sample a driftless normalized spot path starting in $S_1 = 1$ using

$$\bar{dg}_t = \frac{b' z_{t,1} + \sigma_\perp \epsilon}{\sqrt{b'b + \sigma_\perp^2}}, \qquad S_{t+1} = S_t\,(F_t^1)^{-1}\Big(\Phi(\bar{dg}_t)\Big)\ . \tag{29}$$

Because F_t^1 is the option-implied risk-neutral front-expiry marginal, this construction does not estimate or add a physical spot drift.

3.2.2 Time Series Generation

The basic statistical DYSANOS time series generator is summarized as follows:

1. Initialize h_0 from an observed date or from the invariant state distribution.
2. Propagate the state with the exact Gaussian transition (23), or use the tail diffusion methods described in appendix C on page 28.
3. Simulate spot with (29).
4. Form $u_t = b + Bh_t$, map it to bounded SANOS parameters through (14)–(16) and construct the marginal densities as described in section 2.3.

[8] For each date we tabulate F_t^1 on a fine log-return grid and invert it with a monotone PCHIP spline.

5. Evaluate $C_t(T, K)$ at any desired strike and expiry. For a fixed cash contract (τ, k) use $T = \tau - t$ and $K = k/S_t$. Option prices can be pre-computed if their relative strikes and expiries are known in advance; otherwise they can be evaluated on the path on the fly.[9]

3.3 SANOS PCA Implied Volatility

In this section we review the PCA dynamics implied by our model.

Although h_t for our affine-200 model has 200 coordinates, the actual variation is much lower-dimensional. Indeed, the PCA of the AR(1) state process reveals that only a few principal components capture most of the variation in the decoded implied-volatility space as figure 3 and table 4 show. Practically eight PCA vectors are sufficient to drive the entire surface over the time period of 891 dates from 2022-01-03 to 2025-08-29. This is in line with other works [CFD02].

Cumulative decoded-IV variance	Explained variance
PC1	94.62%
PC1–2	97.53%
PC1–4	99.19%
PC1–8	99.76%

Table 4: Decoded implied-volatility variance under the invariant state law.

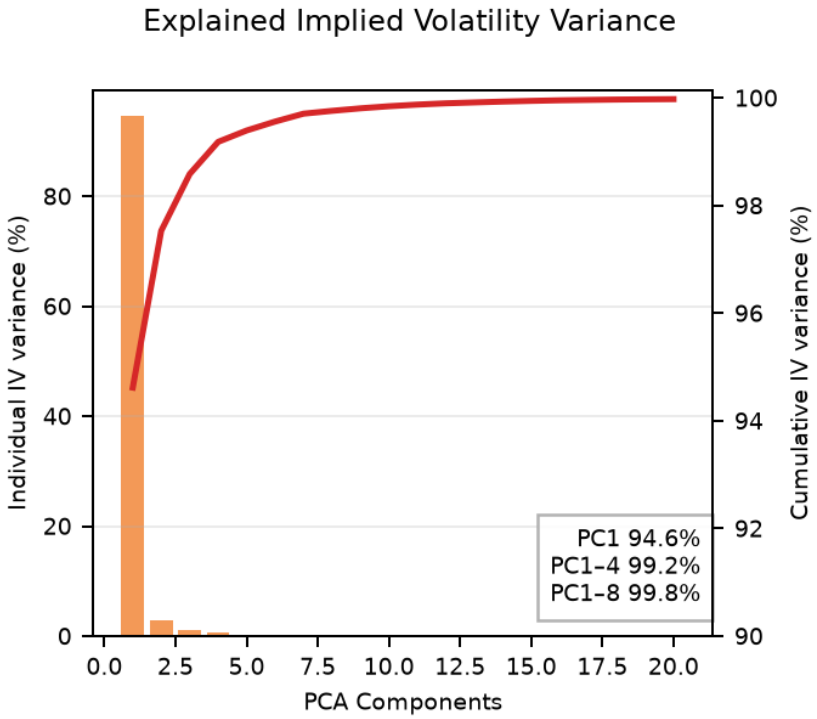


Figure 3: Variance explained by PCA of decoded implied-volatility levels under the fitted invariant state law. The 200 affine coordinates are not themselves these PCs.

Figure 4 shows the first eight eigensurfaces. Each panel is one invariant standard deviation and has its own vertical scale; PCA signs are arbitrary. PC1 is the global level. PC2 is primarily

[9] It is much more efficient if the time to expiry of dynamically generated options is known in advance as then the respective linear expression is local to the surrounding model expiries. If the expiry is entirely free then the price of an arbitrary option in batch-based GPU evaluation is still linear but requires without additional tricks evaluating all model options across the entire density surface.

a broad skew/term-structure rotation, while PC3 couples skew and maturity. Higher modes become increasingly localized, including at the explicit 1D–4D model expiries.

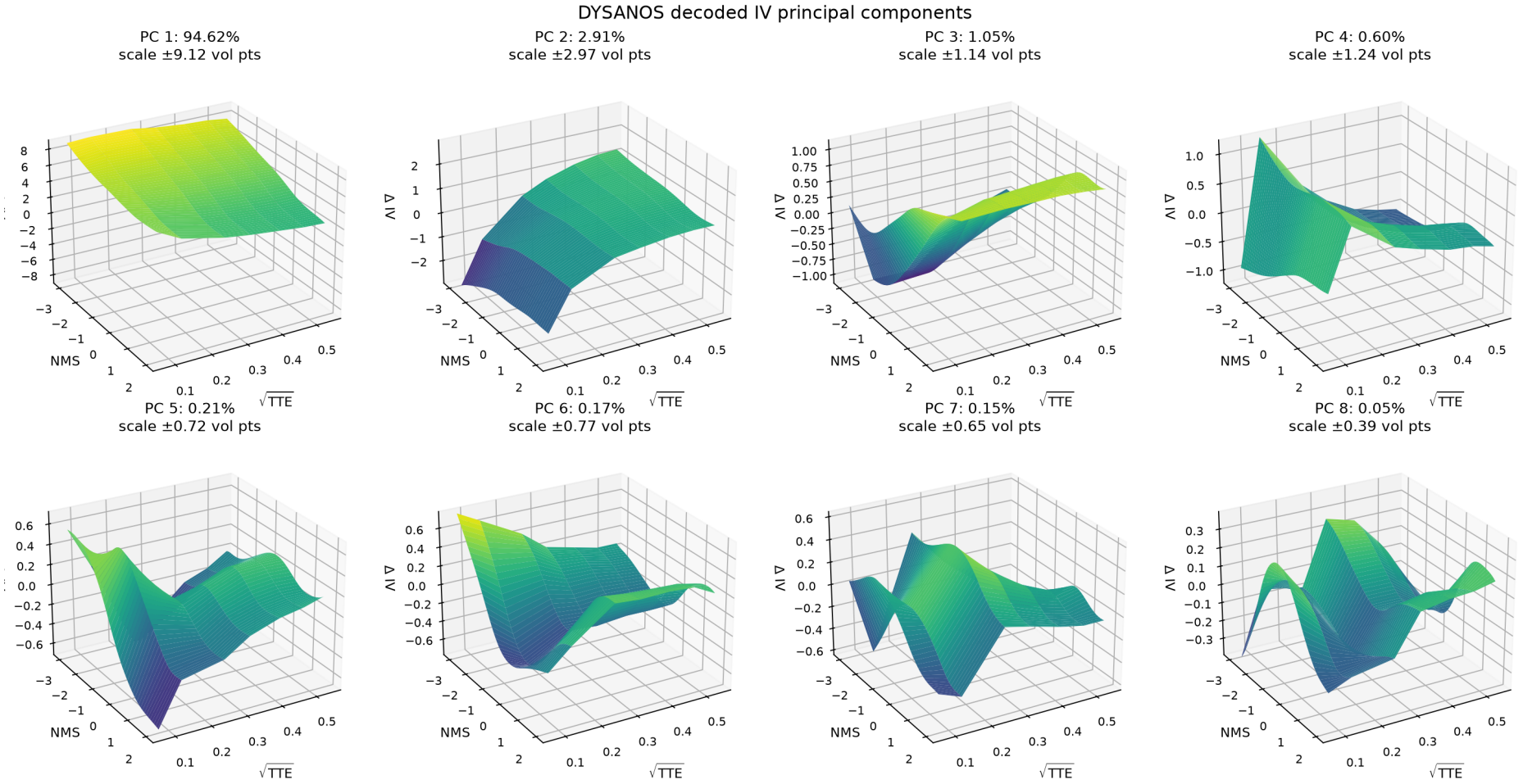


Figure 4: PCA factors 1–8 expressed in implied volatility space as shifts over the mean b of (22).

For completeness we also illustrate the mean b from (22) in figure 5.

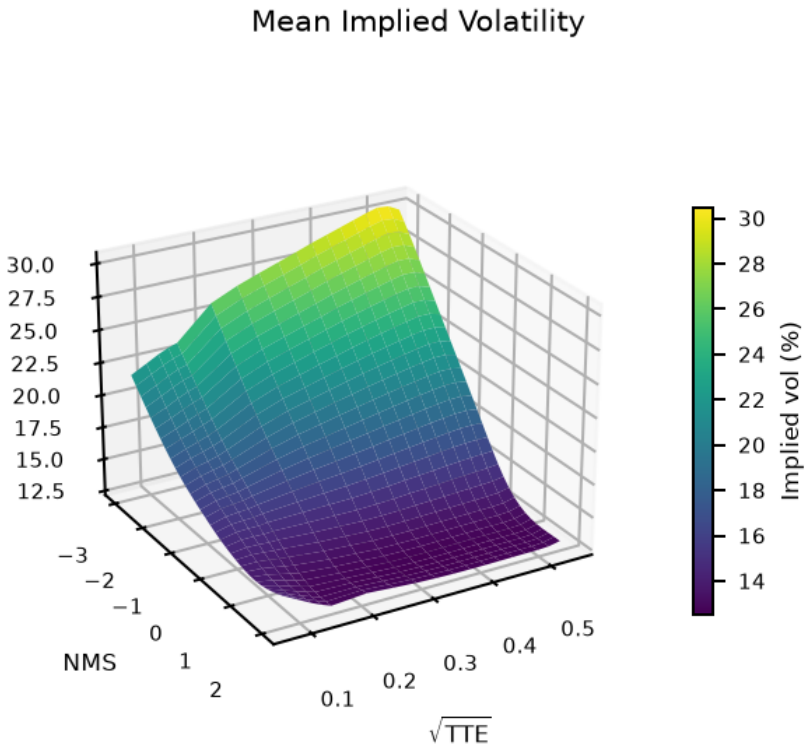


Figure 5: Mean implied-volatility b from (22).

3.3.1 Invariant Law and Stability diagnostics

Statistic	Value
State dimension	200
Dates	891
Exact 1W innovations	861
$\rho(A_5)$	0.9561
Slowest-mode half-life	15.4 weeks
Minimum one-day real eigenvalue of κ	0.00898
Invariant covariance condition number	3.64
Invariant covariance effective rank	196.4

Table 5: Fitted state-law diagnostics.

The law is stable and retains almost full effective rank in state space. A simulator can therefore start from an observed state or draw $h_0 \sim \mathcal{N}(m, P)$ when a stationary initialization is appropriate.

3.3.2 Chronological surface-path comparison

Horizon radii test the latent innovation law but do not directly test paths of decoded option prices. We therefore form every available sequence of four fitted market-state surfaces separated by one full trading week, giving three consecutive one-week changes. Here "observed" means the date-specific latent state fitted to that day's irregular quote book and then passed through the common SANOS decoder; it is not a second interpolation of the raw quotes. Every candidate

simulator starts from that fitted surface at the same origin date. Its path features are changes in normalized call prices on the Cartesian panel of target locations

$$T \in \{1, 5, 10, 20, 40, 70\} \text{ business days}, \qquad K/S \in \{0.85, 0.90, 0.95, 1.00, 1.05, 1.10, 1.15\}.$$

Each target expiry is mapped to the closest distinct model expiry. The model therefore has $3 \times 6 \times 7 = 126$ features. Coordinates are centered and scaled using pre-2025 observed paths only.

For this path-level test, the historical comparator replays an entire pre-2025 three-week sequence of fitted residuals, rather than drawing its three one-week residuals independently. It is therefore a deliberately strong finite-sample benchmark that preserves short-run clustering, but it is not a new parametric law and cannot generate innovations outside the stored blocks.

We report the standardized energy distance and an unbiased Gaussian-kernel maximum mean discrepancy (MMD) between observed and simulated 2025 paths. For the null distribution, entire paired chronological blocks of 15 origin dates swap their observed/simulated labels; the block length matches the three-week feature horizon and retains most overlap-induced dependence. 1,999 random block swaps give a Monte Carlo p-value with minimum positive resolution 0.0005.

The 2025 split is a chronological holdout for fitting any auxiliary tail law, constructing a historical-residual pool, and estimating the feature standardization. The shared affine decoder and the baseline Gaussian AR law are nevertheless taken from the all-history checkpoint. The path comparison is therefore a holdout diagnostic for the innovation-law extensions conditional on the learned representation, not a fully frozen pre-2025 model forecast.

Table 6 reports 144 paired 2025 paths. The p-value is the block-swap test for MMD; energy distance is reported as a second scale-sensitive discrepancy. None of the five candidates is rejected at 5%. This is evidence against a gross path-distribution failure, but not a ranking theorem. Gaussian has the smallest energy distance and MMD. The Tail Diffusion remains statistically plausible while improving the separate tail and persistence diagnostics above.

Innovation law	Energy	$10^3 \times \text{MMD}^2$	p
Gaussian	0.810	1.594	0.704
IID Student-t	0.854	3.788	0.457
Score-driven Student-t	0.838	3.786	0.416
Tail Diffusion	0.854	3.480	0.415
Historical block replay	0.857	6.968	0.345

Table 6: Chronological decoded-surface path diagnostics on 144 paired 2025 origins. Each path contains three consecutive one-week changes; p is based on 1,999 paired 15-origin block swaps.

4 Risk-Neutral DYSANOS

Static arbitrage-freedom of each decoded surface does not by itself impose a martingale restriction on the option prices. We show that a numerical martingale measure can nevertheless be constructed for a subset of tradable contracts by applying the OCE drift-removal construction of [BMPW22] directly to simulated DYSANOS paths. We do not claim risk-neutrality of the full surface; extending the construction to *all* options is ongoing work.

4.1 Universe

We simulate 25 trading dates over $N = 100{,}000$ paths under the learned measure $\mathbb{P}$. The paths are generated antithetically: at every daily transition we draw the state and independent spot innovations for the first half of the block and use their exact negatives for the second half. The two members are then propagated through the same nonlinear decoder. Consequently, the sample contains $N/2 = 50{,}000$ independent antithetic pairs rather than N independent paths.[10]

[10]The antithetic pairing matters when estimating their Monte Carlo uncertainty.

At each date we can trade spot and 12 floating options at each of the four expiries $5D, 10D, 15D, 20D$ struck on the normalized-moneyness grid

$$z \in \{-1.5, -1.3, -1.1, -0.9, -0.7, -0.5, -0.3, -0.1, 0, 0.1, 0.3, 0.5\}.$$

Once entered, each option is held to expiry. Writing Π_t^i for the (spot or option) entry price and X_t^i for the (spot or option) expiry value of instrument i, returns are expressed as fractional gains:

$$R_t^i = \frac{X_t^i}{\Pi_t^i} - 1, \qquad i = 1, \ldots, 49\ . \tag{30}$$

This normalizes our trading to unit \$ amounts invested in each tradable instrument. We only trade ITM options to avoid vanishing premia.

Remark 4.1. *We are going to learn a measure for the* fixed *100,000 paths we have generated. It is not a general function of the state that would generalize to other generated paths. We are looking for an in-sample risk-neutral measure. Therefore, we do not split training and validation set for the below analysis.*

A different task is to find a generic risk-neutral measure for a trained DYSANOS model. This is not the focus of the current analysis.

4.2 Finding a Risk Neutral Measure

As a first step we find a pre-conditioning measure $\mathbb{Q}_1$ equivalent to $\mathbb{P}$ under which the unconditional expectation of every return is zero to numerical tolerance. Let D be the density of $\mathbb{Q}_1$ with respect to $\mathbb{P}$ over $\mathbb{R}^N$. Then we solve[11]

$$\inf_{D \in \mathbb{R}^N} \mathbb{E}^{\mathbb{P}}\left[D \log D\right] \tag{31}$$

subject to the linear constraints

$$D > 0\ , \quad \mathbb{E}^{\mathbb{P}}\left[D\right] = 1 \quad \text{and} \quad \mathbb{E}^{\mathbb{P}}\left[D\, R_t\right] = 0 \tag{32}$$

Table 7 shows summary results of this step.

Diagnostic	Reported value
Simulated paths	100,000
Returns	$1{,}225 = 25 \times 49$
Moment constraints	exact zero
Sample relative entropy	0.2011
Effective sample size	62,877 (62.88% of paths)
Minimum / maximum density	0.0019 / 44.53
Maximum unconditional $\lvert z \rvert$	6.85×10^{-9}

Table 7: Entropy pre-conditioning, evaluated on the 100,000-path sample used in the diagnostics below.

The next step is to construct a measure which has numerically zero conditional expected returns, i.e., a risk-neutral measure. To this end, recall the Minimal Equivalent Martingale Measure (MEMM) [Fri00], which is a special case of our [BMPW22] construction:

[11] Numerically, we solve the smooth unconstrained dual $\inf_a \log[N^{-1}\sum_n \exp(-R_n' a)]$ using trust-region Newton–CG with analytic gradients and exact covariance Hessian–vector products. Float64 log-sum-exp stabilization is used for the objective and gradient, after which $D_n = \exp(-R_n' a)/[N^{-1}\sum_m \exp(-R_m' a)]$ is reconstructed; hence every weight is strictly positive and their sample mean is one.

Theorem 4.2. *Assume our option/spot returns R are $L^1(\mathbb{Q}_1)$ and let $\mathcal{A}$ be the set of bounded variables $a = (a_0, \ldots, a_{24})$ each with values in $\mathbb{R}^{49}$, measurable with respect to the filtration generated by the market. Solve*

$$\sup_{a \in \mathcal{A}} : -\log \mathbb{E}^1 \left[\exp\left(-\sum_{t=0}^{24} \left(a_t' R_t - |a_t|' c_t \right) \right) \right] \longrightarrow a^0 \tag{33}$$

If a^0 is finite, then

$$\frac{d\mathbb{Q}_2}{d\mathbb{Q}_1} := \frac{\exp\left[-\sum_{t=0}^{24} \left(a_t^{0\prime} R_t - |a_t^0|' c_t \right) \right]}{\mathbb{E}^1 \left[\exp\left[-\sum_{t=0}^{24} \left(a_t^{0\prime} R_t - |a_t^0|' c_t \right) \right] \right]} \tag{34}$$

defines an equivalent measure.

In our example, we parameterize the action set a as follows: for expiry k, let $x_{t,k} = (t, h_t, \log \Pi_t^{k,1}, \ldots, \log \Pi_t^{k,12})$. The action network has a two-hidden-layer width-128 SiLU component $F_\theta : \mathbb{R}^{213} \to \mathbb{R}^{13}$ shared across expiries and an expiry-specific affine skip $b_{\theta,k} + W_{\theta,k} x_{t,k}$. Writing

$$r_{\theta,t,k} = b_{\theta,k} + W_{\theta,k} x_{t,k} + F_\theta(x_{t,k}) \in \mathbb{R}^{13}, \qquad k = 1, \ldots, 4, \tag{35}$$

the four first coordinates are summed into one spot action and the remaining coordinates give the 48 option actions. This defines $a_\theta(x_t) \in \mathbb{R}^{49}$ with $\Theta = 56{,}709$ parameters. Thus the strategy uses the full 200-dimensional state, all option entry prices, and calendar time.

Let c_t contain one basis point for spot and ten basis points for every option. The $\mathbb{Q}_2$ fit uses $c_t^{(2)} = c_t/10$, i.e., 0.1 basis points for spot and one basis point for every option, together with the capped action

$$\bar{a}_\theta(x_t) = a_\theta(x_t) \min\left(1, \frac{32}{\|a_\theta(x_t)\|_1}\right) . \tag{36}$$

We then solve the problem

$$\sup_{\theta \in \mathbb{R}^\Theta} : -\log \mathbb{E}^1 \left[\exp\left(-\sum_{t=0}^{24} \left\{ \bar{a}_\theta(x_t)' R_t - |\bar{a}_\theta(x_t)|' c_t^{(2)} \right\} \right) \right] \longrightarrow \theta^0. \tag{37}$$

on the full 100,000 paths, using an exact full-population Adam warm start followed by exact full-population L-BFGS. The full objective and exact-gradient history is reported in Appendix A. Our candidate martingale measure is therefore

$$\frac{d\mathbb{Q}_2}{d\mathbb{Q}_1} = \frac{\exp\left(-\sum_t \left\{ \bar{a}_{\theta^0}(x_t)' R_t - |\bar{a}_{\theta^0}(x_t)|' c_t^{(2)} \right\} \right)}{\mathbb{E}^1 \left[\exp\left(-\sum_t \left\{ \bar{a}_{\theta^0}(x_t)' R_t - |\bar{a}_{\theta^0}(x_t)|' c_t^{(2)} \right\} \right) \right]} . \tag{38}$$

4.3 No-Arb Test

We deploy two types of tests for the no-arbitrage condition: unconditional and conditional.

4.3.1 Unconditional Test

Every unconditional day–instrument mean is compared with its Monte Carlo standard error. We reconstruct the $\mathbb{Q}_1$ and terminal cap-32 $\mathbb{Q}_2$ densities on the same 100,000 paths, split into sets A and B while keeping antithetic pairs together. For each $\mathbb{Q} \in \{\mathbb{P}, \mathbb{Q}_1, \mathbb{Q}_2\}$, write the self-normalised weights as $w_p = d\mathbb{Q}/d\mathbb{P}(p)$. Their effective sample sizes are 100,000, 71,175, and 33,380, respectively. For each of the $25 \times 49 = 1{,}225$ coordinates define

$$\widehat{m}_{t,i} = \frac{\sum_p w_p R_{p,t}^i}{\sum_p w_p}, \qquad z_{t,i} = \frac{\widehat{m}_{t,i}}{\widehat{\mathrm{se}}(\widehat{m}_{t,i})}. \tag{39}$$

The standard error is the self-normalised ratio-estimator influence standard error after averaging the two paths in each antithetic pair. We also report the materiality-regularised score

$$s_{t,i}^{(\varepsilon)} = \frac{\widehat{m}_{t,i}}{\sqrt{\widehat{\mathrm{se}}(\widehat{m}_{t,i})^2 + \varepsilon^2}}, \qquad \varepsilon = 10^{-8}. \tag{40}$$

Diagnostic	$\mathbb{P}$	$\mathbb{Q}_1$	$\mathbb{Q}_2$
Paths / antithetic pairs	100,000 / 50,000	100,000 / 50,000	100,000 / 50,000
Effective sample size	100,000	71,175	33,380
Day–instrument coordinates	1,225	1,225	1,225
Maximum $\lvert z\rvert$	53.026	2.205	2.646
99th percentile $\lvert z\rvert$	45.706	2.032	1.833
$N_{\lvert z\rvert>1.96}$	1,131	19	9
$N_{\lvert z\rvert>2.576}$	1,096	0	1
$N_{\lvert z\rvert>5}$	984	0	0
Maximum $\lvert s^{(\varepsilon)}\rvert$	53.026	2.205	2.646
$N_{\lvert s^{(\varepsilon)}\rvert>1.96}$	1,131	19	9
$N_{\lvert s^{(\varepsilon)}\rvert>2.576}$	1,096	0	1
$N_{\lvert s^{(\varepsilon)}\rvert>5}$	984	0	0

Table 8: Unconditional coordinate diagnostics under $\mathbb{P}$, the pre-conditioned measure $\mathbb{Q}_1$, and the risk-neutral measure $\mathbb{Q}_2$. All three columns use the same paths; antithetic pairs are the independent sampling units.

Under $\mathbb{P}$, 984 coordinates exceed five standard errors; the worst is the day-1 20D $z = -0.7$ ITM call return, with mean -0.0540420, standard error 0.0010192, and $z = -53.026$. Under $\mathbb{Q}_1$, no coordinate exceeds 2.576 standard errors; its worst coordinate is the day-1 spot return, with mean 0.0000636, standard error 0.0000288, and $z = 2.205$. Under $\mathbb{Q}_2$, one coordinate exceeds 2.576 standard errors and none exceeds five; its worst coordinate is the day-4 spot return, with mean -0.0001772, standard error 0.0000670, and $z = -2.646$. Coordinate dependence means that the counts are diagnostics rather than family-wise p-values.

4.3.2 Conditional Test

As a next step, we test whether the conditional returns of our trading instruments have zero mean. This is complicated by the nature of non-recombining Monte-Carlo simulated paths where no state (practically) appears twice.

We therefore propose a novel conditional no-arbitrage test:

Theorem 4.3 (Conditional No-Arb Test)**.** *Let $\mathbb{Q}$ be a measure on a filtered space $(\Omega, \mathcal{F}_0, \ldots, \mathcal{F}_m)$ and let $R_t \in L^1(\mathbb{Q}; \mathbb{R}^d)$, $t = 0, \ldots, m-1$. Let $\mathcal{A}$ contain all bounded $\mathcal{F}_t$-measurable actions a_t. Assume the conditional Laplace transforms used below are finite and twice differentiable on a convex neighbourhood containing zero and the optimizer. If $a^* = (a_0^*, \ldots, a_{m-1}^*)$ is a finite solution of*

$$\inf_{a \in \mathcal{A}} : \mathbb{E}^{\mathbb{Q}}\left[\sum_{t=0}^{m-1} \exp\left(-a_t' R_t + \frac{1}{2} a_t' a_t\right)\right], \tag{41}$$

then, for every t,

$$\mathbb{E}^{\mathbb{Q}}[a_t^{*\prime} R_t] = 0 \quad \Longleftrightarrow \quad \mathbb{E}_t^{\mathbb{Q}}[R_t] = 0 \quad \mathbb{Q}\text{-a.s.} \tag{42}$$

(For a proof refer to the appendix page 29.)

Accordingly, we first split our 100,000 paths into two sets A and B, each containing 50,000 paths (25,000 antithetic pairs). For each $\mathbb{Q} \in \{\mathbb{P}, \mathbb{Q}_1, \mathbb{Q}_2\}$ fit an action with the full affine-plus-deep architecture in (35), but with no action cap, by minimizing[12]

$$\widehat{L}_{A,\mathbb{Q}}(\phi) = \log\left\{ \frac{1}{25} \widehat{\mathbb{E}}_A^{\mathbb{Q}} \left[\sum_{t=0}^{24} \exp\left(-a_\phi(x_t)' R_t + \frac{1}{2} a_\phi(x_t)' a_\phi(x_t) \right) \right] \right\}. \tag{43}$$

For conditionally Gaussian returns, the population action is approximately $(I + \mathrm{Cov}_t(R))^{-1} \mathbb{E}_t^{\mathbb{Q}_j}[R]$.

After fitting on A, the action is evaluated on B using

$$d^{(j)}_{p,A\to B} = \frac{1}{25} \sum_{t=0}^{24} a_{\widehat{\phi}_{A,j}}(x_{p,t})' R_{p,t}. \tag{44}$$

We then exchange A and B and pool the two out-of-fold score vectors. The combined mean is

$$\widehat{m}^{(j)}_{\mathrm{OOF}} = \frac{\sum_{p\in B} w_p d^{(j)}_{p,A\to B} + \sum_{p\in A} w_p d^{(j)}_{p,B\to A}}{\sum_{p\in A\cup B} w_p}. \tag{45}$$

Let $d_p^{(j)}$ denote the pooled held-out score and $\bar{w} = N^{-1} \sum_p w_p$. For an antithetic pair k, define its self-normalised ratio influence by

$$\bar{\psi}^{(j)}_k = \frac{1}{2} \sum_{p\in k} \frac{w_p}{\bar{w}} \left(d^{(j)}_p - \widehat{m}^{(j)}_{\mathrm{OOF}} \right), \qquad \widehat{\mathrm{se}}^{(j)}_{\mathrm{pair}} = \frac{\mathrm{sd}_{k=1,\dots,50{,}000}(\bar{\psi}^{(j)}_k)}{\sqrt{50{,}000}}. \tag{46}$$

The reported action statistic and two-sided normal p-value are

$$T^{(j)}_{\mathrm{act}} = \frac{\widehat{m}^{(j)}_{\mathrm{OOF}}}{\widehat{\mathrm{se}}^{(j)}_{\mathrm{pair}}}, \qquad p^{(j)}_{\mathrm{act}} = 2\Phi\left(- \left| T^{(j)}_{\mathrm{act}} \right| \right) \tag{47}$$

where Φ is the standard-normal distribution function. Under the null of zero conditional drift, $T^{(j)}_{\mathrm{act}}$ is asymptotically standard normal.

Diagnostic	$\mathbb{P}$	$\mathbb{Q}_1$	$\mathbb{Q}_2$
Paths / antithetic pairs	100,000 / 50,000	100,000 / 50,000	100,000 / 50,000
T_{act}	93.414	41.365	−0.454
Two-sided p-value	10^{-1897}	10^{-374}	0.650

Table 9: Conditional zero test, confirming that $\mathbb{Q}_2$ is numerically a martingale measure.

Measure	N	Max. $\lvert z\rvert$	99% $\lvert z\rvert$	$N_{2.576}$	N_5	T_{act}
		Unconditional				
$\mathbb{Q}_1$	1,225	2.205	2.032	0	0	41.365
$\mathbb{Q}_2$	1,225	2.646	1.833	1	0	−0.454

Table 10: Threshold-count summary of the 100,000-path DYSANOS diagnostics. Here N is the number of day–instrument coordinates and $N_c = \#\{|z| > c\}$. Further confirmation that $\mathbb{Q}_2$ is a martingale measure.

The conditional mean test fails clearly for $\mathbb{P}$ and $\mathbb{Q}_1$, but passes comfortably for $\mathbb{Q}_2$.

Remark 4.4. *Solving* (37) *is* slow. *In our examples took several hours on an RTX 5090 to find a measure* $\mathbb{Q}_2$. *That means with the given L-BFGS solver this method is not practically useful if a result is required in minutes or less.*

[12] Features are standardized using the weighted training fold only; both folds and measures use the same 300-epoch schedule: 100 affine-initialization epochs followed by 200 full-network epochs.

4.4 Scope and limitations

This is a reduced demonstration rather than a general construction. The experiment is numerically heavy and the terminal $\mathbb{Q}_2$ policy and conditional actions did not meet strict full-parameter gradient stopping criteria within the fixed 300-epoch two-stage schedule. The frozen-action test remains valid, but incomplete action optimization can reduce its power.

The result is therefore evidence for zero drift within the stated 49-instrument book and strategy class, but it is by construction not a risk-neutrality statement for other strikes, expiries, trading frequencies, or the full DYSANOS surface. The model-free reweighting step also relies on having many more simulated paths than tradable instruments. Extending the construction to the full surface is the natural next step.

5 Conclusion

We presented with DYSANOS the first generative option market model built from smooth, strictly statically arbitrage-free SANOS surfaces. Our approach does not rely on approximate static constraints; each smooth option surface is strictly arbitrage free and allows pricing of any option on GPU.

We have shown how to leverage our representation to build a simple, but already fairly powerful baseline model which can generate spot and option paths for many days in the future up to years. The surface-first AR(1) captures first-order level, skew, term, and leverage effects. Its front and term smoothing coordinates allow the short end to move separately from later expiries, and the affine model places 80.92% of the full quote population inside bid–ask across all 891 historic dates. The Tail Diffusion improves the deficient Gaussian tail law and reproduces tail persistence, although multi-week tail mass remains too low.

For a reduced 49-instrument book on the same simulated paths, the MEMM construction in Section 4 builds a candidate martingale measure $\mathbb{Q}_2$. In the 100,000-path unconditional test the largest coordinate statistic is $|z| = 2.646$, with no coordinate above five. The two-fold uncapped affine-plus-deep test gives $T_{\text{act}} = 41.365$ under $\mathbb{Q}_1$ and $T_{\text{act}} = -0.454$ ($p = 0.650$) under $\mathbb{Q}_2$. We therefore detect no residual conditional drift under $\mathbb{Q}_2$ in the tested book and strategy class. This remains a reduced, computationally heavy demonstration rather than a full-surface risk-neutral construction.

6 Thanks

We want to thank the team at Wharton for their WRDS https://wrds-www.wharton.upenn.edu portal and Option Metrics https://optionmetrics.com/united-states for providing their IvyDB data via this service. Without this data our work would not have been possible.

A Risk-neutral optimizer convergence

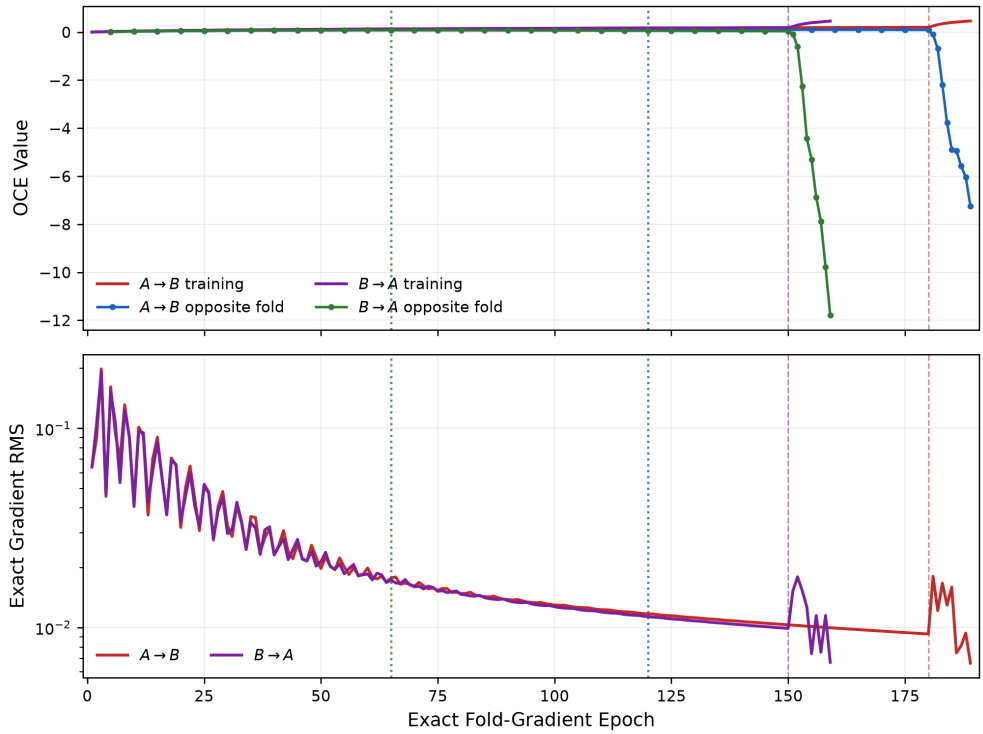


Figure 6: Full-population optimization of (37). The top panel shows the OCE objective and the bottom panel its exact parameter-gradient RMS against cumulative full-population objective/gradient evaluations. The vertical dashed line marks the switch from Adam to L-BFGS; the horizontal dotted line is the terminal stopping target.

B Training the affine state-space model

This appendix provides details on the estimation sequence behind the three rows of model comparisons in Table 3. All stages use the same 891 market dates and equal total weight per date, irrespective of the number of quotes observed on that date.

B.1 Direct surface benchmark

For each date, the direct benchmark optimizes an independent $u_t \in \mathbb{R}^{M(N+1)+2}$ against (20). The raw log-DLV coordinates are unbounded above, whereas pricing mechanically applies the cap in (16). To retain a restoring gradient around and above that cap, define

$$u^{\Sigma}_{\max} := \log(\Sigma_{\max} - \Sigma_{\min}), \qquad a := 0.10, \qquad s := 0.35, \tag{48}$$

and

$$r_t^{j,i} := \frac{a}{s} \log\left[1 + \exp\left(\frac{u_t^{\Sigma,j,i} - u^{\Sigma}_{\max}}{a}\right)\right]. \tag{49}$$

The full-data cap penalty is

$$\mathcal{L}_{\text{cap}} := \frac{1}{D} \sum_{t=1}^{D} \frac{1}{2} \sum_{j=1}^{M} \sum_{i=1}^{N} (r_t^{j,i})^2. \tag{50}$$

Far above the cap this approaches a quadratic penalty in the excess raw log-DLV coordinate with scale s; a determines the smooth transition. The direct benchmark minimizes the equal-date quote loss plus $\mathcal{L}_{\text{cap}}$.

B.2 Static affine-200 dimension reduction

The direct states $u_t^\star$ provide only an initialization for the affine model. An analytic rank-200 singular-value decomposition first solves the least-squares reconstruction problem

$$\min_{b,B,\{h_t\}} \sum_{t=1}^{D} \|u_t^\star - (b + Bh_t)\|_2^2.$$

The resulting b, B, and h_t are then fine-tuned directly against the market objective

$$\frac{1}{D}\sum_{t=1}^{D} \mathcal{L}_t^{\text{quote}}(b + Bh_t) + \mathcal{L}_{\text{cap}}. \tag{51}$$

The reconstruction loss to $u_t^\star$ is not retained during this phase. The static affine-200 results in Table 3 are therefore market-objective fits, not untouched PCA reconstructions. This phase contains no AR likelihood.

B.3 Exact affine re-gauging and AR initialization

Before fitting dynamics, the completed affine coordinates are centered and whitened by an exact change of coordinates. The decoder bias and loading are transformed at the same time so that every decoded state $u_t = b + Bh_t$ is unchanged to numerical precision. Hence this operation changes neither a surface nor a fitted option price.

For the re-gauged states, every origin date whose endpoint lies exactly five NYSE sessions later supplies one pair (h_t, h_s). There are 861 such overlapping pairs. Ordinary least squares supplies an initial five-session transition A_5. Its first-order generator proxy is rescaled to per-session units; the symmetric part is projected onto the dissipative cone while its skew-symmetric part is retained. The initial innovation covariance is the population covariance of the resulting one-week residuals, with a $10^{-6}I$ numerical ridge. Its upper Cholesky factor initializes G_5. This produces a stable warm start for (23); it is not yet the final joint fit.

B.4 Joint AR overlay

The joint stage runs for 300 epochs and alternates two kinds of optimizer steps.

1. The *surface steps* traverse all dates in shuffled batches and update b, B, and the corresponding h_t using the quote loss and DLV-cap penalty.

2. One *transition step* uses all 861 exact-five-session pairs and updates h_t, m, the dissipative generator parameters, and the full upper-triangular innovation factor G_5 using $\mathcal{L}_{\text{AR}}$ and the gauge penalty below. The decoder b, B receives no direct gradient from this transition step.

The dynamics weight is ramped linearly over the first 50 epochs. The state, decoder, and dynamics learning rates start at 10^{-4}, 3×10^{-5}, and 3×10^{-4} respectively, after a 20-epoch warm-up, and follow a cosine schedule to one tenth of those values. Gradient norms are capped at 10.

For $\bar{h} = D^{-1}\sum_t h_t$, define the population covariance

$$\widehat{\Gamma}_h := \frac{1}{D}\sum_{t=1}^{D}(h_t - \bar{h})(h_t - \bar{h})'. \tag{52}$$

The affine-gauge penalty is

$$\mathcal{L}_{\text{gauge}} := \frac{1}{d}\|\bar{h}\|_2^2 + \frac{1}{d}\|\widehat{\Gamma}_h - I_d\|_F^2. \tag{53}$$

It controls translations, scale, and shear of the otherwise non-identifiable affine coordinates; orthogonal rotations and reflections remain immaterial. The joint stage uses the weights stated below (27). All 891 dates enter both its surface and transition objectives, so its fit and dynamics diagnostics are in-sample.

The unrestricted direct state was not given an analogous full-covariance joint AR overlay. Its dimension is 3,620, exceeding the 861 observed one-week transitions. A nonsingular unrestricted innovation covariance is therefore not identified without another dimension-reduction or covariance model.

C Appendix: Tail Dynamics

This section provides alternatives to the Gaussian diffusion of $h_{t+\Delta}$ in (23) aimed to better capture tail behavior.

Gaussian innovations are a convenient baseline but miss market tails. As a non-parametric comparator we also simulate at one-week steps by independently resampling centered fitted one-week residuals $e_{t,s}$ from (25) whose endpoints precede 2025. At two and three weeks, independently drawn residuals are propagated through A and summed; no multi-week block is replayed in this tail experiment.

For an exact-horizon state return r with innovation factor G_H, define its normalized squared radius

$$(\delta^H)^2 := \frac{\left\|(G_H')^{-1} r\right\|_2^2}{200}. \tag{54}$$

The Gaussian reference would concentrate this quantity sharply around one in 200 dimensions. The observed radii are both heavier-tailed and serially dependent. Table 11 makes the scale of the discrepancy explicit.

Innovation law	1W	2W	3W
Gaussian	0.03%	$< 0.01\%$	$< 0.01\%$
Independent historical replay	5.15%	2.41%	2.32%
Historical/Gaussian ratio	162.01×	17,197.36×	17,430.29×

Table 11: Probability of exceeding the pre-2025 empirical 95% normalized-radius threshold under Gaussian innovations and independent historical resampling. Ratios use the unrounded probabilities; each simulated estimate uses one million paths.

To obtain a more realistic tail law, we add a scalar latent log-variance diffusion without replaying any historical residual. Keeping the affine conditional mean and loading G from the fitted factor G_5, the exact one-week transition is

$$\varepsilon_{n+1} = \sqrt{\rho \exp(g_n - v_g/2)}\, G_5' z_{n+1}, \qquad z_{n+1} \sim \mathcal{N}(0, I), \tag{55}$$

$$g_{n+1} = \phi g_n + \sqrt{v_g(1-\phi^2)}\, \xi_{n+1}, \qquad \xi_{n+1} \sim \mathcal{N}(0,1), \tag{56}$$

with independent z_{n+1} and ξ_{n+1}. Stationarity gives $g_n \sim \mathcal{N}(0, v_g)$ and hence $\mathbb{E}[\rho \exp(g_n - v_g/2)] = \rho$. Equation (56) is the exact discrete transition of a stationary Ornstein–Uhlenbeck diffusion; it therefore permits fractional-time simulation without Euler discretization.

The parameters (ρ, ϕ, v_g) are fitted on the 703 pre-2025 one-week innovations. Since

$$\log(\delta_n^2) = \log \rho + g_n - v_g/2 + \log(\chi^2_{200}/200), \tag{57}$$

we use a Kalman likelihood whose measurement mean and variance are the exact first two moments of $\log(\chi^2_{200}/200)$; only its Gaussian shape is an approximation. The fit gives $(\rho, \phi, v_g) =$

$(0.878, 0.139, 0.0649)$. On the 158 2025 innovations, log-radius negative log likelihood falls from 2.453 to 0.125 relative to a pre-2025-scaled Gaussian benchmark.

For comparison we also fit variance-normalized multivariate Student-t innovations, either IID or with a persistent score-driven common scale. The fitted degrees of freedom are 34.5. On the same 2025 innovations the score-driven law reduces full-vector negative log likelihood per dimension from 1.351 to 1.332; the improvement over IID Student-t is negligible.

Table 12 reports the probability of exceeding the pre-2025 empirical 95% radius at each horizon. "Market" is the realized 2025 rate; every simulated estimate uses one million paths.

Innovation law	1W	2W	3W
2025 market	6.33%	6.33%	5.70%
Gaussian	0.03%	$< 0.01\%$	$< 0.01\%$
IID Student-t	4.14%	0.99%	0.86%
Score-driven Student-t	4.12%	1.07%	0.93%
Tail Diffusion	3.75%	0.78%	0.66%
Independent historical replay	5.15%	2.41%	2.32%

Table 12: Probability of exceeding the pre-2025 market 95% normalized-radius threshold. Market frequencies use 158 2025 transitions; simulation uses one million paths per law.

The Tail Diffusion is an improvement over the Gaussian law. It produces a squared-radius serial correlation of 0.116, compared with 0.091 in the market history; Gaussian and IID laws have zero population correlation. It has, however, somewhat less tail mass than IID Student-t at every horizon.[13] While independent historical replay is still closest in marginal tail mass, the Tail Diffusion is a fully parametric generator.

D Proof of The Conditional No Arb Test

Proof of theorem 4.3. Because the objective is additive over t and $\mathcal{A}$ contains all bounded adapted actions, a_t^* minimizes the corresponding conditional objective pointwise. For a fixed state define

$$\ell_t(u) := \frac{1}{2} u'u + \log \mathbb{E}_t^{\mathbb{Q}}[\exp(-u'R_t)]. \tag{58}$$

Let $\widetilde{\mathbb{Q}}_{t,u}$ denote the conditional law obtained by tilting $\mathbb{Q}(\,\cdot \mid \mathcal{F}_t)$ proportionally to $\exp(-u'R_t)$. Then

$$\nabla \ell_t(u) = u - \mathbb{E}_t^{\widetilde{\mathbb{Q}}_{t,u}}[R_t], \qquad \nabla^2 \ell_t(u) = I + \mathrm{Cov}_t^{\widetilde{\mathbb{Q}}_{t,u}}(R_t) \succeq I. \tag{59}$$

Thus ℓ_t is strictly convex, its minimizer is unique, and $\nabla \ell_t(0) = -\mu_t$ for $\mu_t := \mathbb{E}_t^{\mathbb{Q}}[R_t]$. Strong monotonicity of its gradient gives

$$a_t^{*\prime} \mu_t = (\nabla \ell_t(a_t^*) - \nabla \ell_t(0))' a_t^* \geq \|a_t^*\|^2, \tag{60}$$

where $\nabla \ell_t(a_t^*) = 0$. Consequently,

$$\mathbb{E}^{\mathbb{Q}}[a_t^{*\prime} R_t] = \mathbb{E}^{\mathbb{Q}}[a_t^{*\prime} \mu_t] \geq \mathbb{E}^{\mathbb{Q}}[\|a_t^*\|^2]. \tag{61}$$

If $\mu_t = 0$, strict convexity makes $a_t^* = 0$, and the captured return is zero. Conversely, if the captured return is zero, the last display implies $a_t^* = 0$ almost surely; its first-order condition then gives $\mu_t = 0$ almost surely. □

[13] The 2025 sample is small and overlapping: 15-origin circular-block bootstrap 95% intervals for the market exceedance rates are approximately 0.6–14.6% across the three horizons. Differences of a percentage point should therefore not be over-interpreted.

E Appendix: Data Pipeline

In the interest of full transparency we here discuss main elements of the data pipeline used in our experiments.

The market-data pipeline constructs a standardized daily time series of SPX option surfaces. Raw option quotes are cleaned, normalized, filtered, checked against an arbitrage-free reference surface, and finally collected into irregular quote books suitable for direct model training.

The pipeline processes SPX option data between January 2nd, 2020 and August 29th, 2025. The latter is the latest day available at the time of writing. Only valid full-day NYSE trading days are considered.

E.1 Pipeline Overview

The complete transformation can be summarized as

$$\begin{aligned}\text{raw SPX option quotes} &\longrightarrow \text{daily cleaning and enrichment}\\ &\longrightarrow \text{arbitrage-free quote surface using linear SANOS}\\ &\longrightarrow \text{liquidity and moneyness filters}\\ &\longrightarrow \text{shared 200-strike SANOS model geometry}\\ &\longrightarrow \text{direct native and affine-200 quote fits.}\end{aligned}$$

The key insight when working with Option Metrics IvyDB data is that the forward curves from the `Forward_Price` table do not line up with the option data themselves. We therefore compute our own forward from weighted put-call parities. We also convert the tightest options into call prices. Such data still has arbitrage, hence we use a linear SANOS pre-processing step to obtain fitted prices within bid/ask which adhere to basic linear no-arbitrage conditions (the absence of which would indicate actionable arbitrage). We drop any strikes where such a fit cannot be achieved, assuming that fit issues are due to data alignment challenges rather than genuine trading opportunities.

E.2 Raw Daily Market Data

For every eligible date d, the underlying data contain quantities such as

- the SPX spot price S_d and previous spot price S_{d-1};
- option strikes and expiration dates; call and put bid and ask quotes; volume and open interest; AM/PM settlement indicators;
- forward and discount-curve information (with forwards being re-calculated below).

The default raw-data cleaning removes

- same-day expirations;
- options with bids below 0.011 (the minimum tick size is 0.05\$ for series trading below 3\$);
- observations with missing or invalid quotes; observations with an invalid settlement flag; irregular expirations; dates or ticker-date combinations known to contain bad data (e.g. 2025-01-08 which was a holiday).

E.3 Enrichment and Expiry Filtering

The cleaned quotes are enriched with

- business days and year fractions to expiry where AM options are counted as 20% of the day;
- implied forward prices computed off put-call parity per expiry for all strikes for which both call and put were traded and have valid bid/ask prices and sizes, weighted by put-call implied forward bid-ask spread.[14]
- using the new forward, define "market" mid call prices between the actual call prices and synthetic call prices from the puts;
- compute implied volatilities; forward- and discount-adjusted prices, strikes, and Greeks.

An expiration is retained only if it satisfies the default liquidity and quality requirements. In particular,

- at least 10 strikes must have either a traded call or put;
- at least 4 strikes must have both a traded call and put;
- the discounted forward normalized spread must be at least 10^{-6};
- implied volatility must lie in
$$0.01 \leq \sigma^{\text{impl}} \leq 5.$$

E.4 Arbitrage-Free Preprocessing

Before the final option filter is applied, an arbitrage-free representation of the daily market is constructed using [BHK+26]. The default configuration uses surface mode and performs the following operations:

1. Extend the strike grid to improve boundary treatment.
2. Fit a linear (DLV) no-arbitrage option-price curve separately for each expiry.
3. Drop expiries for which a valid fit cannot be obtained.[15]
4. Remove locally redundant strikes associated with negligible fitted density.
5. Fit a surface across strike and expiry to control both butterfly and calendar arbitrage.

At least five strikes per expiry are required. The default minimum fitted density is 10^{-4}, and the minimum admissible ATM volatility is 0.05.

[14] We compute $\text{fwd}^{\text{ask}}(K) := +\text{Call}^{\text{ask}} - \text{Put}^{\text{bid}} - \text{df}K$ and $\text{fwd}^{\text{bid}}(K) := +\text{Call}^{\text{bid}} - \text{Put}^{\text{ask}} - \text{df}K$ using the discount factor sourced from IvyDB. Then we define $\text{fwd}^{\text{mid}}(K) := 0.5\,(\text{fwd}^{\text{ask}}(K) + \text{fwd}^{\text{bid}}(K))$ and $\gamma^{\text{mid}}(K) := 0.5\,(\text{fwd}^{\text{ask}}(K) - \text{fwd}^{\text{bid}}(K))$ and
$$\text{fwd} := \frac{\sum_K \gamma(K)^{-1}\text{fwd}^{\text{mid}}(K)}{\sum_K \gamma(K)^{-1}}$$

[15] Existence of such a fit is equivalent to the market not having actionable arbitrage using bid/asks as quoted.

E.5 Option Filtering

The empirical analysis retains observations satisfying

$$\text{BDTE} \leq 70,$$

where BDTE denotes business days to expiry.

The normalized-moneyness filter is

$$-3 \leq z_{\text{ATM}} \leq 2,$$

where z_{ATM} is normalized moneyness computed using the arbitrage-free fitted ATM volatility for the corresponding expiry.

Additional requirements are

$$\begin{aligned} \text{VegaSqrtVar} &\geq 10^{-3}, \\ \log(\text{volume}) &\geq 1, \\ \log(\text{open interest}) &\geq 1. \end{aligned}$$

Here, "VegaSqrtVar" refers to Vega divided by sqrt of expiry. It is therefore the sensitivity to total volatility of the option.

Each retained expiry must contain at least 10 options.

It should be noted that this filter removes many options post 1Y for the early years of the training period due to the minimum volume traded filter. We suspect that this information in IvyDB is not reliable in the past. Future tests might want to remove the volume and/or open interest filters. (It is an option in our data pipeline parametrization.)

E.6 Spot Normalization

Price-like quantities are normalized by the current spot S_d to obtain variables of scale around 1. For example,

$$\widetilde{K}_d^i = \frac{K_d^i}{S_d}, \qquad \widetilde{C}_d^i = \frac{C_d^i}{S_d}, \qquad \widetilde{F}_d^i = \frac{F_d^i}{S_d}.$$

The original spot is retained as $\text{spot_ref}_d = S_d$. (The normalized log-spot level is passed on as h^0 later.)

E.7 Arbitrage-free preprocessing and model geometry

The preceding steps produce tradable normalized calls and puts with their observed bid–ask intervals. A per-date arbitrage-free SANOS fit supplies a smooth reference surface, an ATM-volatility curve, and an availability check for the retained quotes. The completed Direct-200 experiments do *not* resample this information onto a secondary reference grid. Instead, every eligible original quote enters the loss at its observed strike and expiry.

The decoder uses the 18 model expiries stated in Section 3: 1D–5D followed by 10D, 15D, and five-session increments through 70D. On each date it builds one shared support with $N = 200$ inner strikes. Half of the normalized-moneyness candidates cover the first model expiry and half cover the final model expiry over $z \in (-3, 2)$. The candidates are merged and spaced, and the two boundary strikes are

$$\hat{K}_t^0 = \tfrac{1}{2}\hat{K}_t^1, \qquad \hat{K}_t^{N+1} = \tfrac{3}{2}\hat{K}_t^N.$$

The same support is used at every model expiry on that date; only the density and DLV coordinates carry an expiry superscript.

E.8 Fitting ML-SANOS

The native control first fits one unrestricted vector $u_t \in \mathbb{R}^{M(N+1)+2}$ to each date's irregular quote book. The completed affine model then uses the map

$$h_t \in \mathbb{R}^{200} \longmapsto u_t = b + Bh_t \in \mathbb{R}^{M(N+1)+2}$$

from (22). It is initialized from the native controls and first fine-tuned across all 891 dates using only the market objective. The stable AR overlay then jointly refines this affine model and its one-week transition. Prices are evaluated directly at every observed quote; the training weights and inside-spread objective are those in (20). Appendix B describes the complete optimization sequence.

References

[AH11] J. Andreasen and B. Huge. Volatility interpolation. *Risk*, pages 86–89, 2011.

[BHK+26] Hans Buehler, Blanka Horvath, Anastais Kratsios, Yannick Limmer, and Raeid Saqur. Sanos - smooth arbitrage-free non-parametric option surfaces. https://arxiv.org/abs/2601.11209, Februar 2026.

[BMPW22] Hans Buehler, Phillip Murray, Mikko S. Pakkanen, and Ben Wood. Deep hedging: learning to remove the drift. *Risk*, February 2022.

[BR15] Hans Buehler and Evgeny Ryskin. Discrete local volatility for large time steps (extended version). https://papers.ssrn.com/abstract=2642630, 2015.

[Bue06] Hans Buehler. Expensive martingales. *Quantitative Finance*, 6(3):207–218, 2006.

[Bue08] Hans Buehler. Volatility and dividends - volatility modelling with cash dividends and simple credit risk. https://papers.ssrn.com/abstract=1141877, 2008.

[CFD02] Rama Cont, Jose da Fonseca, and Valdo Durrleman. Stochastic models of implied volatility surfaces. *Economic Notes*, 31(2):361–377, 2002.

[CM99] Peter P. Carr and Dilip B. Madan. Option valuation using the fast fourier transform. *Journal of Computational Finance*, 2(4):61–73, 1999.

[CN12] R. Carmona and S. Nadtochiy. Tangent levy market models. *Finance and Stochastics*, 16:63–104, 2012.

[CRW23] Samuel N. Cohen, Christoph Reisinger, and Sheng Wang. Arbitrage-free neural-sde market models. *Applied Mathematical Finance*, 30(1):1–46, 2023.

[Fen09] Matthias R. Fengler. Arbitrage-free smoothing of the implied volatility surface. *Quantitative Finance*, 9(4):417–428, 2009.

[Fri00] Marco Frittelli. The minimal entropy martingale measure and the valuation problem in incomplete markets. *Mathematical Finance*, 10(1):39–52, 2000.

[GHL12] Julien Guyon and Pierre Henry-Labordère. Being particular about calibration. *Risk*, January 2012.

[NJZB23] Brian Ning, Sebastian Jaimungal, Xiaorong Zhang, and Maxime Bergeron. Arbitrage-free implied volatility surface generation with variational autoencoders. *SIAM Journal on Financial Mathematics*, 14(4):1004–1027, 2023.

[WBWB19] Magnus Wiese, Lianjun Bai, Ben Wood, and Hans Buehler. Deep hedging: Learning to simulate equity option markets. https://arxiv.org/abs/1911.01700, 2019.

[WWP+21] Magnus Wiese, Ben Wood, Alexandre Pachoud, Ralf Korn, Hans Buehler, Phillip Murray, and Lianjun Bai. Multi-asset spot and option market simulation. https://arxiv.org/abs/2112.06823, 2021.